

1. Suppose that a book of 300 pages contains 200 misprints. Use Poisson approximation to write down the probability that there is more than one misprint on a particular page.

$$\text{misprints per page on average} = \frac{200}{300} = \frac{2}{3} \quad \alpha = \frac{2}{3}$$

$$\text{Poisson distribution} = a_k = \frac{\alpha^k e^{-\alpha}}{k!}$$

X - random variable for the misprints on a page

$$\begin{aligned} P(X > 1) &= 1 - [P(X=0) + P(X=1)] \\ &= 1 - \left[\left(\frac{\alpha^0 e^{-\alpha}}{0!} \right) + \left(\frac{\alpha^1 e^{-\alpha}}{1!} \right) \right] \\ &= 1 - \left(\frac{\alpha^0 e^{-\alpha}}{0!} + \frac{\alpha^1 e^{-\alpha}}{1!} \right) \\ &= 1 - \left(e^{-\frac{2}{3}} + \frac{2}{3} e^{-\frac{2}{3}} \right) \\ &= 1 - \frac{5}{3} e^{-\frac{2}{3}} \end{aligned}$$

2. In a school where 4% of the children write with their left hands, what is the probability that there are no left-handed children in a class of 25?

$$P(\text{lefty}) = 0.04 \quad \left(1 - \frac{4}{100}\right)^{25} \approx e^{-1}$$

3. Six dice are thrown 200 times by the players. Estimate the probability of obtaining "six different faces" k times, where $k = 0, 1, 2, 3, 4, 5$.

$$P(\text{six different faces}) = \frac{6!}{6^6}$$

$$\alpha = \frac{6!}{6^6} (200) = \frac{250}{81}$$

$$\text{Poisson distribution: } a_k = \frac{\alpha^k e^{-\alpha}}{k!}$$

$$\alpha = \frac{250}{81} \quad k = 0, 1, 2, 3, 4, 5$$

6. Find the maximum term or terms in the binomial distribution $B_k(n; p)$, $0 \leq k \leq n$. Show that the terms increase up to the maximum and then decrease. [Hint: take ratios of consecutive terms.]

ratios of consecutive terms:

$$\frac{\binom{n}{k} p^k (1-p)^{n-k}}{\binom{n}{k-1} p^{k-1} (1-p)^{n-k+1}} = \frac{(n-k+1)p}{k(1-p)}$$

For decreasing,

$$\frac{(n-k+1)p}{k(1-p)} < 1$$

$$np - kp + p < k - kp$$

$$k > (n+1)p$$

For increasing,

$$\frac{(n-k+1)p}{k(1-p)} > 1$$

$$(n+1)p > k$$

The binomial distribution terms increases and then decreases.

If $(n+1)p$ is not an integer, the maximum term of binomial distribution is at $k = [(n+1)p]$, $[x]$ denotes the greatest integer not exceeding x .

If $k = (n+1)p$ is an integer, the maximum terms are $k = (n+1)p$ and $k = (n+1)p - 1$

7. Find the maximum term or terms in the Poisson distribution $\pi_k(\alpha)$, $0 \leq k < \infty$. Show the same behavior of the terms as in No. 6.

$$\begin{aligned} \text{Ratio of consecutive terms} &= \frac{P(k)}{P(k-1)} = \frac{\alpha^k e^{-\alpha}}{k!} \cdot \frac{(k-1)!}{\alpha^{k-1} e^{-\alpha}} \\ &= \frac{\alpha}{k}, \quad k = 0, 1, 2, 3, \dots \end{aligned}$$

$$\begin{cases} \text{when } k \leq \alpha, P(k) \geq P(k-1) \\ \text{when } k > \alpha, P(k) < P(k-1) \end{cases}$$

If α is not an integer, the maximum term of binomial distribution is at $k = [\alpha]$.

If α is integer, the max terms are at $k = \alpha - 1$ & $k = \alpha$

8. Let X be a random variable such that $P(X = c + kh) = \pi_k(\alpha)$, where c is a real and h is a positive number. Find the Laplace transform of X .

$$P(X = c + kh) = \pi_k(\alpha) = \frac{\alpha^k e^{-\alpha}}{k!}$$

$$g(z) = E(z^X)$$

$$\text{let } z = e^{-\lambda}$$

$$g(e^{-\lambda}) = E(e^{-\lambda X}) = \alpha(\lambda)$$

$$\alpha(\lambda) = \sum_{k=0}^{\infty} P(X = c + kh) e^{-\lambda(c + kh)}$$

$$= \sum_{k=0}^{\infty} \frac{\alpha^k e^{-\alpha}}{k!} e^{-\lambda c} \cdot e^{-\lambda k h}$$

$$= e^{-\alpha - \lambda c} \sum_{k=0}^{\infty} \frac{(\alpha e^{-\lambda h})^k}{k!}$$

$$= e^{-\alpha - \lambda c} \cdot e^{\alpha e^{-\lambda h}} = \exp[-\lambda c + \alpha(e^{-\lambda h} - 1)]$$

9. Find the convolution of two sequences given by Poisson distributions $\{\pi_k(\alpha)\}$ and $\{\pi_k(\beta)\}$.

$$\text{let } i \sim P(\alpha), \quad j \sim P(\beta)$$

$$\pi_k(\alpha) = \frac{\alpha^i e^{-\alpha}}{i!}, \quad i = 0, 1, 2, 3, \dots$$

$$\pi_k(\beta) = \frac{\beta^j e^{-\beta}}{j!}, \quad j = 0, 1, 2, 3, \dots$$

$$P(I + J = k) = \sum_{n=0}^k P(I = n, J = k - n)$$

i and j are independent

$$P(I + J = k) = \sum_{n=0}^k \left(\frac{\alpha^n e^{-\alpha}}{n!} \right) \left(\frac{\beta^{k-n} e^{-\beta}}{(k-n)!} \right)$$

$$= e^{-(\alpha + \beta)} \sum_{n=0}^k \frac{\alpha^n \beta^{k-n}}{n! (k-n)!}$$

$$= e^{-(\alpha + \beta)} (\alpha + \beta)^k \quad k = 0, 1, 2, 3, \dots \text{ is Poisson } (\alpha + \beta) = \pi_k(\alpha + \beta)$$

奖励题1

1) Let $Y = \int_{-\infty}^{\infty} e^{-(i\xi+x)^2} dx$ let $i\xi+x = z$ $dx=dz$ when $x \rightarrow \infty, z \rightarrow \infty$
 let $i\xi+x = a$ $dx=da$ when $x \rightarrow -\infty, z \rightarrow -\infty$

$$Y = \int_{-\infty}^{\infty} e^{-z^2} dz$$

$$Y^2 = Y \cdot Y = \int_{-\infty}^{\infty} e^{-z^2} dz \cdot \int_{-\infty}^{\infty} e^{-a^2} da$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(z^2+a^2)} dz da$$

$$= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r dr d\theta$$

convert to polar coordinates by Jacobian:

$$z^2 + a^2 = r^2 \quad z = r \cos \theta \quad a = r \sin \theta$$

$$J = \frac{\partial(z, a)}{\partial(r, \theta)} = r \cos^2 \theta + r \sin^2 \theta = r$$

$$\therefore dz da = r dr d\theta$$

$$\text{let } r^2 = t$$

$$\frac{dt}{dr} = -2r$$

$$dt = -2r dr$$

$$\frac{dt}{-2} = r dr$$

$$= \int_0^{2\pi} \left[\int_0^{\infty} e^{-t} \left(\frac{dt}{-2} \right) \right] d\theta$$

$$= -\frac{1}{2} \int_0^{2\pi} e^{-t} \Big|_0^{\infty} d\theta$$

$$= -\frac{1}{2} \int_0^{2\pi} -1 d\theta$$

$$= \frac{1}{2} (2\pi)$$

$$= \pi$$

$$Y^2 = \pi \quad Y = \sqrt{\pi} \Rightarrow \int_{-\infty}^{\infty} e^{-(i\xi+x)^2} dx = \sqrt{\pi} \quad \Delta Q.E.D.$$

2) $X \sim B(n, p)$, $Y \sim B(m, p)$ are independent

their moment generating function $M_X(t) = (q + pe^t)^n$

$$M_Y(t) = (q + pe^t)^m$$

$$M_{(X+Y)}(t) = E[e^{t(X+Y)}]$$

$$= E[e^{tx}] \cdot E[e^{ty}]$$

$$= M_X(t) \cdot M_Y(t)$$

$$M_{(X+Y)}(t) = M_X(t) \cdot M_Y(t)$$

$$= (q + pe^t)^n \cdot (q + pe^t)^m$$

$$M_{(X+Y)}(t) = (q + pe^t)^{n+m}$$

$$\therefore X+Y \sim B(n+m, p)$$